DATANOVA 2026 · DN-CY-01

NetSentinel

Finding rare network attacks, including the ones nobody has seen yet

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THE PROBLEM

Millions of normal connections. A handful of attacks.

Attacks are rare. In the real traffic used here, there are **45,000 normal flows** for every flow of the rarest attack.

That breaks naive machine learning: a detector that calls everything normal looks accurate and catches nothing.

80%

accuracy of a detector that always answers “normal”

0%

of attacks caught by that same detector

THE APPROACH

Four hard parts, four answers

Attacks are rare

Rebalanced training shows rare attacks more often (SMOTE).

New attacks appear

A second detector learns only normal traffic and flags the unfamiliar.

Tactics change

A drift monitor notices the shift without labels and triggers retraining.

Analysts drown in alerts

Alerts become explained incidents; analyst decisions retrain the model.

HOW IT WORKS

From raw traffic to an analyst’s decision

01
Flow records
One row per connection, any dataset format

→

02
Clean and audit
Duplicates, broken rows and leaky columns removed

→

03
Two detectors
Known attacks: rebalanced ensemble. New ones: novelty detector

→

04
Triage
Incidents with priority, reason and first response

→

05
Analyst
Confirms or rejects; the model retrains on it

One command runs the whole pipeline and writes a full evaluation report: **python -m nids all**

REAL TRAFFIC

Tested on CIC-IDS2017

2.8 M

network flows recorded over five working days

8

attack families, from DDoS to Heartbleed, plus normal traffic

45,000:1

normal flows per flow of the rarest attack (sample used)

Real data is messy. Repaired automatically: a 12-hour clock with no AM/PM, 288,602 empty rows, 308,218 duplicate flows and a broken label encoding.

RESULTS · 121,196 TEST FLOWS

Known attack types: almost all caught

99.9%

of attack flows raise an alert

0.10%

of normal flows raise a false alarm

0.973

macro-F1, every family weighted equally (0.766 without rebalancing)

Honest reading: most of the jump from 0.766 comes from 9 test flows of the two rarest families (Heartbleed 2 of 2 found, Infiltration 5 of 7, versus none). On families with at least 10 test flows, rebalancing adds +0.005: small, but statistically significant.

THE HONEST NUMBER

A supervised detector only knows yesterday's attacks



Every attack in CIC-IDS2017 happens on a single day. Trained on Monday to Thursday morning, the model has never seen Friday's DDoS, port scans or botnet. That is why NetSentinel adds a novelty detector and fast retraining.

UNSEEN ATTACKS

Catching attacks it has never seen

Hidden from training	Classifier alone	With the novelty detector
Web attacks	3%	**87%**
DoS	3%	**71%**
DDoS	24%	**40%**
Brute force	0%	**3%**
Port scans, botnet	0%	**0%: look normal at flow level**

Each family is hidden from training in turn. Average share of its flows that still raise an alert: **5%** → **33%**, at about 1% extra false alarms.

CHANGING TRAFFIC

When traffic changes, it notices and recovers

0.998

on past traffic, Monday to Thursday morning

→

0.182

on the following days, model not updated

→

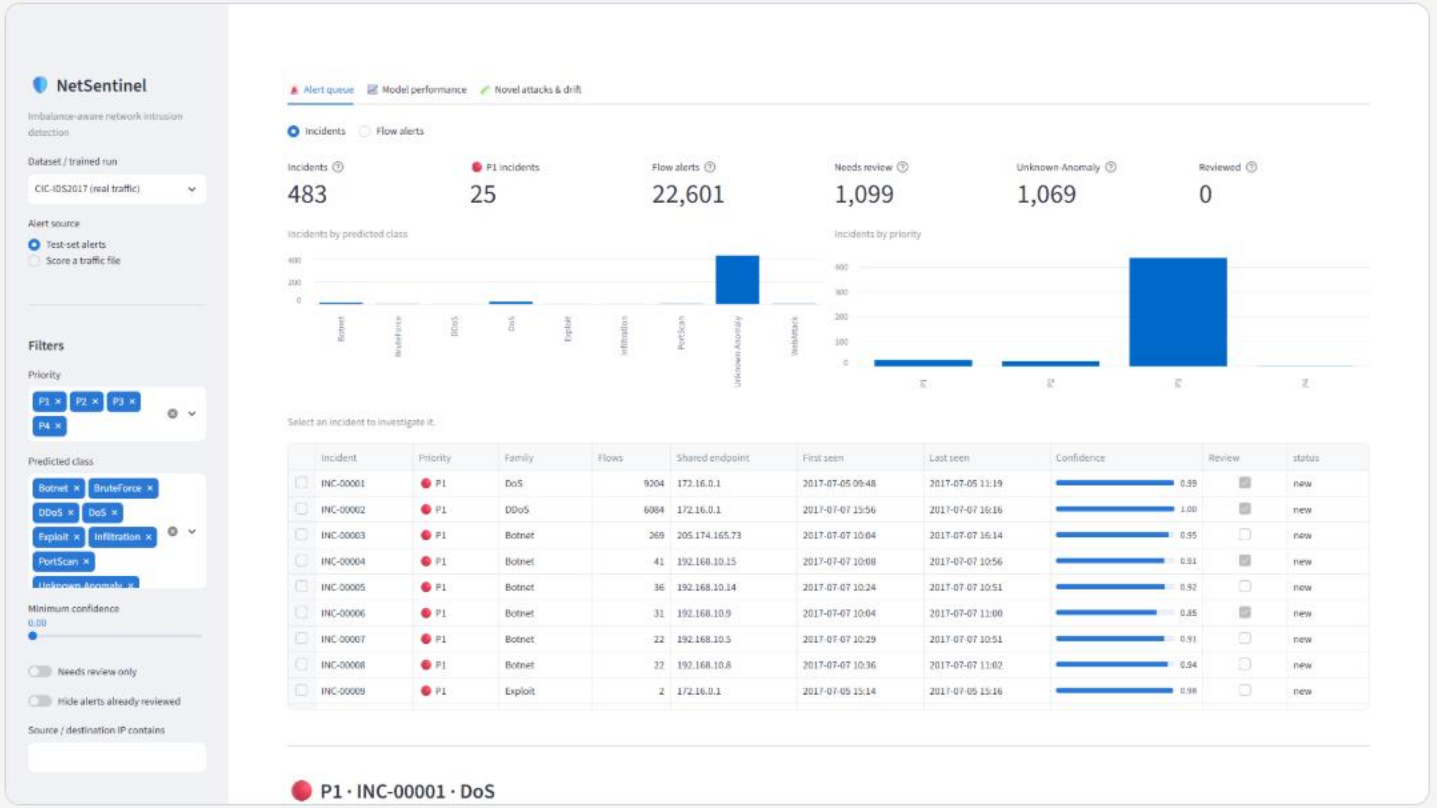
0.925

after retraining on 10% labelled new flows

The drift monitor fired **without any labels**: the alerted traffic shifted, the mix of predictions shifted, and the share of “unknown” anomalies rose from 0.9% to 5.4%.

ANALYST CONSOLE

From 22,601 alerts to 483 incidents



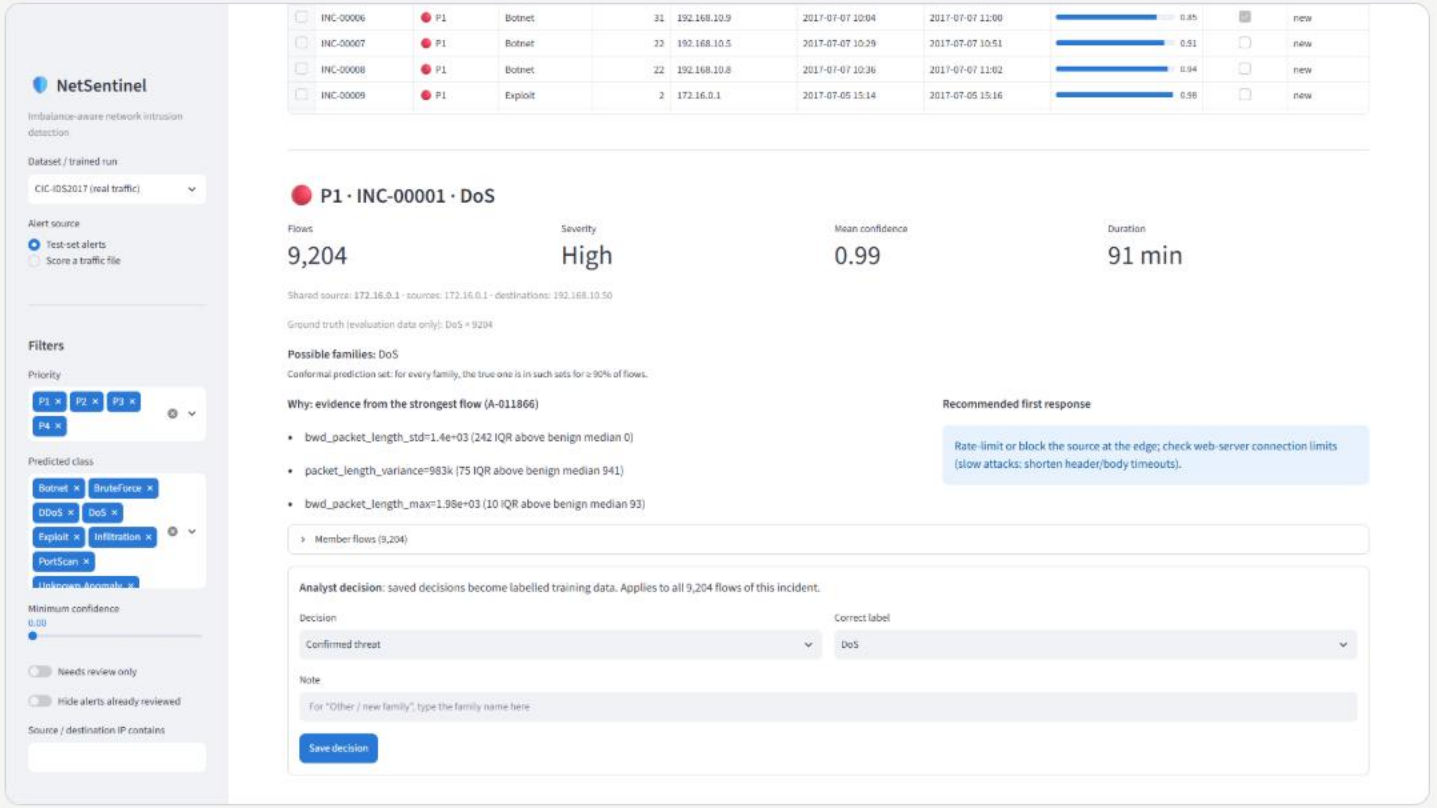
47×

fewer items to open: one incident per attack family, shared endpoint and time window

Priority P1–P4 from severity and confidence

ANALYST CONSOLE

Every incident explains itself



Why

the features furthest from normal traffic

How sure

possible families, with a 90% guarantee per family (conformal prediction)

What to do

a recommended first response

The loop

one decision labels the whole incident; the model retrains on it

METHODOLOGY

Numbers you can check

Confidence intervals

On every headline number; tiny families shown as plain counts.

Three test protocols

Random, session-grouped and time-based splits, side by side.

Automatic data audit

Shortcut scan, duplicate removal and leakage guards first.

Reproducible

One config, one command, 81 automated tests, open code.

LIMITS AND NEXT STEPS

What it cannot do yet

Limits

- Port scans and botnets look normal at flow level
- The novelty detector costs about 1% false alarms
- CIC-IDS2017 is from 2017 and has known label noise

Next steps

- Pair with payload inspection (Zeek, Suricata)
- Tune the alert budget to the team’s capacity
- Re-run on the official DataNova data: one command

NETSENTINEL

Rare attacks found. New ones flagged. Analysts in the loop.

Code, report and demo: github.com/wpersonxdd/netsentinel-ids

Thank you · Rustamjonov Komron Davronovich · DataNova 2026, DN-CY-01